The relativistic spin group $SL(2,\mathbb{C})$.

Def^n [$SL(2,\mathbb{C})$]

- Set: $SL(2,\mathbb{C}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{C}^4 \;\middle|\; ad - cb = 1 \right\} \subseteq \mathbb{C}^4$.
- Group: $(SL(2,\mathbb{C}), \circ)$ $\quad \circ : SL(2,\mathbb{C}) \times SL(2,\mathbb{C}) \longrightarrow SL(2,\mathbb{C})$.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \circ \begin{pmatrix} e & f \\ g & h \end{pmatrix} := \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}.$$

Check : Closed ?

$$\Rightarrow (ae+bg)(cf+dh) - (ce+dg)(af+bh) \overset{?}{=} 1.$$

$$\Rightarrow [aecf + bgdh] + \{aedh + bgcf\} - [ceaf + dgbh] - \{cebh + dgaf\}$$

$$\Rightarrow ad(eh-gf) + bc(gf-eh) = ad - bc = 1.$$

using: $ad - bc = 1$, $eh - gf = 1$.

⊙ the binary opr $\circ$ is associative.

⊙ ∃ identity element, $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in SL(2,\mathbb{C})$.

⊙ $\forall \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{A} \in SL(2,\mathbb{C}) : \exists \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1}}_{A^{-1}} \in SL(2,\mathbb{C}) : A \circ A^{-1} = A^{-1} \circ A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

$$\longrightarrow \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

$\Rightarrow (SL(2,\mathbb{C}), \circ)$ is a group.

- topological space: $(SL(2,\mathbb{C}), \Theta)$ → for the most part, we'll not construct the actual topology, we'll generate it from some preexisting one (in this case $\mathbb{C}^4$ with standard top. is a obs. choice).

Detour [Topologies on $\mathbb{C}^n$].

– def^n top. $\Theta_{\mathbb{C}}$ on $\mathbb{C}$ by virtue of def^n open balls

$$B_r(z) := \{ y \in \mathbb{C} \mid |y - z| < r \} \; ; \; r \in \mathbb{R}^+ , \; z \in \mathbb{C}.$$

$U \in \Theta_{\mathbb{C}}$ if $\forall p \in U : \exists r \in \mathbb{R}^+ : B_r(p) \subseteq U$.

– on finite Cartesian product $\mathbb{C}^n$, we can define product top. $\Theta_{\underbrace{\mathbb{C}\times\mathbb{C}\times\ldots\times\mathbb{C}}_{n\text{-times}}}$

– $SL(2,\mathbb{C}) \subset \mathbb{C}^4 \rightsquigarrow \Theta := \Theta_{\mathbb{C}\times\mathbb{C}\times\mathbb{C}\times\mathbb{C}}\big|_{SL(2,\mathbb{C})} := \{ U \cap SL(2,\mathbb{C}) \mid U \in \Theta_{\mathbb{C}^4} \}$.

$\Rightarrow (SL(2,\mathbb{C}), \Theta)$ is a topological space.

- $(SL(2,\mathbb{C}), \Theta)$ is a topological manifold. (paracompact, Hausdorff and locally $\mathbb{C}^n$).

– Hausdorff : $\mathbb{C}^4 \cong \mathbb{R}^8$; $\mathbb{R}^n$ is Hausdorff

$\Rightarrow \mathbb{C}^4$ is Hausdorff. $\Rightarrow M \neq N \in \mathbb{C}^4$; $r = \frac{1}{2} d(M,N)$ and $B_r(M) \cap B_r(N) = \emptyset$.

if $M \neq N \in SL(2,\mathbb{C}) \Rightarrow [B_r(M) \cap SL(2,\mathbb{C})] \cap [B_r(N) \cap SL(2,\mathbb{C})] = \emptyset$.

- Paracompact: $\mathbb{C}^4$ is metric space. $\rightsquigarrow$ $\mathcal{O}_{\mathbb{C}^4} = \mathcal{O}_d$

$\Rightarrow SL(2,\mathbb{C})$ metrizable $\Rightarrow$ Paracompact [Stone's Thm].

- Locally $\mathbb{C}^n$: The whole space $\subseteq \mathbb{C}^4$ with one-condition $(ad-bc=1)$ it seems like it has to be $\mathbb{C}^3$.

if we construct explict charts, we are done for local structure.

$$U = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{C}) \;\middle|\; a \neq 0 \right\}$$

first prove U is open. consider

$$f_1\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = a. \qquad SL(2,\mathbb{C}) \xrightarrow{f_1} \mathbb{C}$$

$$\pi_1((a,b,c,d)) = a \qquad \mathbb{C}^4 \xrightarrow{\pi_1} \mathbb{C}$$

π_1 is continuous. $B_r(z) \subset \mathbb{C}$

$$f_1 = \pi_1\big|_{SL(2,\mathbb{C})}$$

$$\pi_1^{-1}(B_r(z)) = \underbrace{B_r(z)}_{\text{open}} \times \underbrace{\mathbb{C}^3}_{\text{open}}$$

open in $\mathbb{C}^4$

$\Rightarrow f_1$ is cts.

$\{0\}$ is closed in $\mathbb{C}$.

$\Rightarrow \mathbb{C}^* = \mathbb{C}\setminus\{0\}$ is open in $\mathbb{C}$.

$\Rightarrow \underbrace{f_1^{-1}(\mathbb{C}^*)}_{U}$ is open in $SL(2,\mathbb{C})$.

Define the coordinate map $x : U \longrightarrow \mathbb{C}^3$.

$$x\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = (a,b,c). \quad \} \text{ cts.}$$

since $a \neq 0 \Rightarrow x(U) = \underbrace{\mathbb{C}^* \times \mathbb{C} \times \mathbb{C}}_{\text{open in } \mathbb{C}^3}$.

$$x^{-1} : x(U) \longrightarrow U \quad \} \text{ cts}$$

$$x^{-1}(a,b,c) = \begin{pmatrix} a & b \\ c & \frac{1+bc}{a} \end{pmatrix} \in SL(2,\mathbb{C}).$$

$\Rightarrow x$ is homeomorphism. (U,x) is a chart on $SL(2,\mathbb{C})$.

Atleast we need one more chart to cover entire set, $SL(2,\mathbb{C})$.

$$V = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{C}) \;\middle|\; b \neq 0 \right\} \; ; \; y\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = (a,b,d) \in \mathbb{C}^3$$

$$y^{-1}(a,b,d) = \begin{pmatrix} a & b \\ \frac{ad-1}{b} & d \end{pmatrix}.$$

$\Rightarrow (V,y)$ is another chart of $SL(2,\mathbb{C})$.

$\circledast$ Notice $U \cup V = SL(2,\mathbb{C})$ it is obs $U \cup V \subseteq SL(2,\mathbb{C})$, we need to prove that $SL(2,\mathbb{C}) \subseteq U \cup V$.

consider $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{C}) \Rightarrow ad - bc = 1$

if $a = 0$ or $a \neq 0$ } two possible cases.

↳ $a \neq 0$

$\Rightarrow A \in U \Rightarrow A \in U \cup V \Rightarrow SL(2,\mathbb{C}) \subseteq U \cup V.$

↳ $a = 0$

$\Rightarrow -bc = 1 \Rightarrow bc = -1 \Rightarrow b \neq 0$

$\Rightarrow A \in V \Rightarrow A \in U \cup V \Rightarrow SL(2,\mathbb{C}) \subseteq U \cup V.$

$\Rightarrow SL(2,\mathbb{C}) \subseteq U \cup V. \Rightarrow U \cup V = SL(2,\mathbb{C}).$

$\Rightarrow \mathcal{A} = \{(U,x), (V,y)\}$ is an atlas on $SL(2,\mathbb{C})$.

$\Rightarrow SL(2,\mathbb{C})$ is complex 3-dimensional topological manifold.

$$\boxed{\dim_{\mathbb{C}} SL(2,\mathbb{C}) = 3.}$$

- $SL(2,\mathbb{C})$ is $\mathbb{C}$-diff'ble manifold.

$\mathcal{A} = \{(U,x), (V,y)\}$ is an atlas.

Check compatibility for all transition maps.

↳ $x \circ y^{-1} : y(U \cap V) \longrightarrow x(U \cap V)$

$\mathbb{C}^* \times \mathbb{C}^* \times \mathbb{C}$ $\qquad$ $\mathbb{C}^* \times \mathbb{C}^* \times \mathbb{C}$

$$(x \circ y^{-1})(a,b,d) = x\begin{pmatrix} a & b \\ \frac{ad-1}{b} & d \end{pmatrix} = \left(a, b, \frac{ad-1}{b}\right)$$

$\Rightarrow \mathbb{C}$-diff'ble.

↳ $y \circ x^{-1} : x(U \cap V) \longrightarrow y(U \cap V).$

$$(y \circ x^{-1})(a,b,c) = y\begin{pmatrix} a & b \\ c & \frac{1+bc}{a} \end{pmatrix} = \left(a, b, \frac{1+bc}{a}\right)$$

$\Rightarrow \mathbb{C}$-diff.

$\Rightarrow \mathcal{A}$ is $\mathbb{C}$-diff'ble atlas.

It contains all the charts which are $\mathbb{C}$-compatible with (U,x) & (V,y)

$\mathcal{A}^m$ be the maximal extension of $\mathcal{A}$, such that $\mathcal{A} \subseteq \mathcal{A}^m$ and $\mathcal{A}^m$ is $\mathbb{C}$-diff atlas.

$\Rightarrow (SL(2,\mathbb{C}), \mathcal{O}, \mathcal{A}^m)$ is a $\mathbb{C}$-diff mfd.

- $SL(2,\mathbb{C})$ is Lie Group.

$$\mu : SL(2,\mathbb{C}) \times SL(2,\mathbb{C}) \longrightarrow SL(2,\mathbb{C})$$

$$\mu\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \begin{pmatrix} e & f \\ g & h \end{pmatrix}\right) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \circ \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

$$\iota : SL(2,\mathbb{C}) \longrightarrow SL(2,\mathbb{C})$$

$$\iota\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Need to show that μ & ι are smooth maps.

↳ focus on ι,

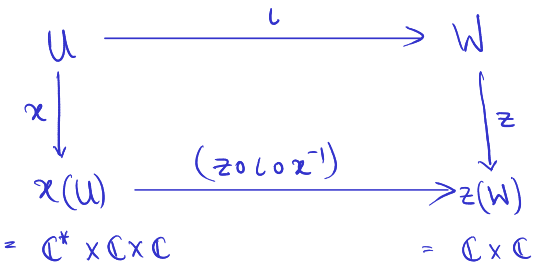
$\iota : SL(2,\mathbb{C}) \longrightarrow SL(2,\mathbb{C})$; if we prove that for U & V the are smooth then we are done.

$$\iota\left(\underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\in U \ (a \neq 0)}\right) = \underbrace{\frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}}_{\in W}$$

$U \xrightarrow{\iota} V.$

def" $W = \{\ldots \mid d \neq 0\}$

$$z\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = (b,c,d)$$



need to show $(y \circ \iota \circ x^{-1})$ is $\mathbb{C}$-diff

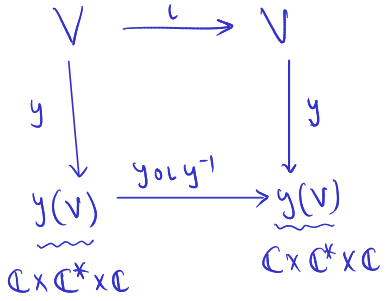
$$(z\circ\iota\circ x^{-1})(a,b,c) = (z\circ\iota)\left(\begin{pmatrix} a & b \\ c & \frac{1+bc}{a}\end{pmatrix}\right) \qquad a\neq 0$$

$$= z\left(\frac{1}{(1+bc)-bc}\begin{pmatrix} \frac{1+bc}{a} & -b \\ -c & a\end{pmatrix}\right)$$

$$= z\begin{pmatrix} \frac{1+bc}{a} & -b \\ -c & a\end{pmatrix}$$

$$= (-b, -c, a). \Rightarrow \text{diff.}$$

Similarly for V.



$$(y\circ\iota\circ y^{-1})(a,b,d) = (y\circ\iota)\begin{pmatrix} a & b \\ \frac{ad-1}{b} & d\end{pmatrix}$$

$$= y\begin{pmatrix} d & -b \\ \frac{1-ad}{b} & a\end{pmatrix}$$

$$= (d, -b, a).$$

$\Rightarrow$ diff.

for any $A \in SL(2,\mathbb{C})$, choose U if $a\neq 0$ or V if $a=0$ & $b\neq 0$ for oper nbd. and we are done for ι is smooth.

⊛ Similarly, $\mu : SL(2,\mathbb{C}) \times SL(2,\mathbb{C}) \longrightarrow SL(2,\mathbb{C})$ is smooth.

$\Rightarrow (SL(2,\mathbb{C}), \circ)$ is complex 3-dimensional Lie group.

- Lie algebra $L(SL(2,\mathbb{C}))$ of Lie group $SL(2,\mathbb{C})$.

Notation: $\mathfrak{sl}(2,\mathbb{C}) \equiv L(SL(2,\mathbb{C}))$.

$$\mathfrak{sl}(2,\mathbb{C}) := \left\{ X \in \Gamma(TSL(2,\mathbb{C})) \mid \forall g \ (l_g)_{*h}(X_h) = X_{g\cdot h} \right\}$$

"left invariant vector field"

we also proved

$$\mathfrak{sl}(2,\mathbb{C}) \cong_{vec} T_{(1,0)}SL(2,\mathbb{C}). \quad \text{and}$$

defined Lie bracket for ↗ using commutator of vector fields.

$$[\![A, B]\!] := j^{-1}\left([j(A), j(B)]\right) \quad ; \quad j : T_{id}G \longrightarrow \Gamma(TG).$$

To get complete Lie algebra, we can get Lie brackets of basis and then use bilinearity to get all of the Lie brackets. For the basis we'll use coordinate induced ones.

To actually compute the Lie bracket, we need to first compute the vector field.

$X \in T_{id} SL(2,\mathbb{C})$

$$X = x^i \underbrace{\left(\frac{\partial}{\partial x^i}\right)_{id}}_{\text{coordinate induced basis}}$$

need chart containing $id = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $\Rightarrow (U, x)$ works.

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{C})$.

$$\underbrace{(l_A)_* \left(\left(\frac{\partial}{\partial x^i}\right)_{id}\right)}_{\text{vector at } A \in T_A SL(2,\mathbb{C})} = j\left(\left(\frac{\partial}{\partial x^i}\right)_{id}\right)_A \qquad \forall\ A \in SL(2,\mathbb{C})$$

$$(l_A)_* \left(\left(\frac{\partial}{\partial x^i}\right)_{id}\right)(f) = \left(\frac{\partial}{\partial x^i}\right)_{id} (f \circ l_A)$$

$$= \partial_i \left(f \circ l_A \circ x^{-1}\right)(x(id))$$

$$= \partial_i \left(\underbrace{f \circ x^{-1}}_{\substack{x(U) \to \mathbb{C} \\ \subseteq \mathbb{C}^3}} \circ \underbrace{x \circ l_A \circ x^{-1}}_{\substack{x(U) \to x(U) \\ \mathbb{C}^3 \to \mathbb{C}^3}}\right)(x(id))$$

chain rule

$$= \partial_m \left(f \circ x^{-1}\right)_{(x \circ l_A \circ x^{-1})(x(id))} \cdot \partial_i \left(x \circ l_A \circ x^{-1}\right)^m (x(id))$$

(m-th comp.)

$(x \circ l_A \circ x^{-1})(x(id)) = (x \circ l_A)(id) = x(A \cdot id) = x(A)$.

$$= \partial_m \left(f \circ x^{-1}\right)_{x(A)} \cdot \partial_i \left(x \circ l_A \circ x^{-1}\right)^m \Big|_{x(id)}$$

$\partial_m (f \circ x^{-1})_{x(A)}$: from def^n of basis tangent vector at A.

$$= \left(\frac{\partial}{\partial x^m}\right)_A (f) \cdot \partial_i \left(x \circ l_A \circ x^{-1}\right)^m \Big|_{x(id)}$$

$\left(\frac{\partial}{\partial x^m}\right)_A (f)$: We got our tangent vector @ A.

$\partial_i (x \circ l_A \circ x^{-1})^m |_{x(id)}$: coefficient for linear sum.

⊛ $x \circ l_A \circ x^{-1} : x(U) \to x(U)$

$$(x \circ l_A \circ x^{-1})(e, f, g) = (x \circ l_A) \begin{pmatrix} e & f \\ g & \frac{1+fg}{e} \end{pmatrix} = x\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} e & f \\ g & \frac{1+fg}{e} \end{pmatrix}\right)$$

only place where group op^r enters the calc.

$$\Rightarrow \quad x\left(\begin{pmatrix} ae+bg & af+\frac{b(1+fg)}{e} \\ ce+dg & cf+\frac{d(1+fg)}{e} \end{pmatrix}\right) = \left(ae+bg,\ af+\frac{b(1+fg)}{e},\ ce+dg\right)$$

we can arrange all these ρ based on (m,i) choice in an array $\begin{bmatrix} \quad \end{bmatrix}$ (m ← row, i ← column)

$$\partial_i\left(x^m \circ l_A \circ x^{-1}\right)(e,f,g) = \begin{bmatrix} a & 0 & b \\ -b/e^2 & a+bg/e & bf/e \\ c & 0 & d \end{bmatrix}$$

Now for actual answer we need $\partial_i\left(x \circ l_A \circ x^{-1}\right)^m\left(\underbrace{x(id)}_{=(1,0,0)}\right)$

$$\partial_i\left(x^m \circ l_A \circ x^{-1}\right)(x(id)) = \begin{bmatrix} a & 0 & b \\ -b & a & 0 \\ c & 0 & d \end{bmatrix} \quad \text{here} \quad d = \frac{1+bc}{a},$$

$$\equiv D^m{}_i(A).$$

$$\Rightarrow j\left(\left(\frac{\partial}{\partial x^i}\right)_{id}\right)_A = (l_A)_*\left(\left(\frac{\partial}{\partial x^i}\right)_{id}\right) = D^m{}_i(A)\left(\frac{\partial}{\partial x^m}\right)_A \in T_A SL(2,\mathbb{C}).$$

$$\Rightarrow \quad j\left(\left(\frac{\partial}{\partial x^i}\right)_{id}\right) = D^m{}_i\left(\frac{\partial}{\partial x^m}\right) \in \Gamma\left(TSL(2,\mathbb{C})\right).$$

$$D^m{}_i = \begin{bmatrix} x^1 & 0 & x^2 \\ -x^2 & x^1 & 0 \\ x^3 & 0 & \frac{1+x_2x_3}{x_1} \end{bmatrix};$$

Need to actually compute Lie Brackets for that, we start by computing commutators of vector fields of kind $j\left(\left(\frac{\partial}{\partial x^i}\right)_{id}\right)$.

$$\left[j\left(\left(\frac{\partial}{\partial x^i}\right)_{id}\right), j\left(\left(\frac{\partial}{\partial x^k}\right)_{id}\right)\right] = \left[D^m{}_i \frac{\partial}{\partial x^m}, D^n{}_k \frac{\partial}{\partial x^n}\right]$$

To compute RHS, $f \in C^\infty\left(SL(2,\mathbb{C})\right)$.

$$\left[D^m{}_i\left(\frac{\partial}{\partial x^m}\right), D^n{}_k\left(\frac{\partial}{\partial x^n}\right)\right](f) = \left(\quad\right)(f).$$

Consider

$$D^m{}_i \frac{\partial}{\partial x^m}\left(D^n{}_k \left(\frac{\partial}{\partial x^n}\right)(f)\right) = D^m{}_i \frac{\partial}{\partial x^m}\left(D^n{}_k \, \partial_n (f\circ x^{-1}) \circ x\right)$$

$= D^m{}_i \frac{\partial}{\partial x^m}\left(D^n{}_k \, \partial_n (f\circ x^{-1}) \circ x\right)$ — $D^n{}_k$: fxn of $SL(2,\mathbb{C})$

$= D^m{}_i \frac{\partial}{\partial x^m}\left(D^n{}_k \, \partial_n (f\circ x^{-1}) \circ x\right)$ — $D^n{}_k$: fxn of $SL(2,\mathbb{C})$

Using Leibniz rule

$$= D^m{}_i \frac{\partial}{\partial x^m}(D^n{}_k)\cdot\left(\partial_n (f\circ x^{-1})\circ x\right) + D^m{}_i D^n{}_k \frac{\partial}{\partial x^m}\left(\partial_n (f\circ x^{-1})\circ x\right)$$

$$= D^m{}_i \frac{\partial}{\partial x^m}(D^n{}_k)\cdot\frac{\partial}{\partial x^n}(f) + D^m{}_i D^n{}_k \left(\partial_m\left(\left[\partial_n (f\circ x^{-1})\circ x\right]\circ x^{-1}\right)\right)\circ x$$

$$= D^m{}_i \frac{\partial}{\partial x^m}(D^n{}_k)\cdot\frac{\partial}{\partial x^n}(f) + D^m{}_i D^n{}_k \left[\partial_m \partial_n (f\circ x^{-1})\right]\circ x$$

Similarly,

$$D^n{}_k \frac{\partial}{\partial x^n}\left(D^m{}_i \frac{\partial}{\partial x^m}(f)\right) = D^n{}_k \frac{\partial}{\partial x^n}(D^m{}_i)\cdot\frac{\partial}{\partial x^m}(f) + D^n{}_k D^m{}_i \left[\partial_n \partial_m (f\circ x^{-1})\right]\circ x$$

$$\Rightarrow \left[D^m{}_i \frac{\partial}{\partial x^m}, D^n{}_k \frac{\partial}{\partial x^n}\right](f) = D^m{}_i \frac{\partial}{\partial x^m}(D^n{}_k)\cdot\frac{\partial}{\partial x^n}(f) - D^n{}_k \frac{\partial}{\partial x^n}(D^m{}_i)\cdot\frac{\partial}{\partial x^m}(f)$$

($m \leftrightarrow n$)

$$= \left[D^m{}_i \frac{\partial}{\partial x^m}(D^n{}_k) - D^m{}_k \frac{\partial}{\partial x^m}(D^n{}_i)\right]\cdot\frac{\partial}{\partial x^n}(f)$$

$$\Rightarrow \left[D^m{}_i \frac{\partial}{\partial x^m}, D^n{}_k \frac{\partial}{\partial x^n}\right] = \left(D^m{}_i \frac{\partial}{\partial x^m}(D^n{}_k) - D^m{}_k \frac{\partial}{\partial x^m}(D^n{}_i)\right)\cdot\frac{\partial}{\partial x^n}$$

$$\left[\left[\left(\frac{\partial}{\partial x^i}\right)_{id}, \left(\frac{\partial}{\partial x^k}\right)_{id}\right]\right] = j^{-1}\left(\left[j\left(\left(\frac{\partial}{\partial x^i}\right)_{id}\right), j\left(\left(\frac{\partial}{\partial x^k}\right)_{id}\right)\right]\right.$$

$$= j^{-1}\left(\left[D^m{}_i \frac{\partial}{\partial x^m}, D^n{}_k \frac{\partial}{\partial x^n}\right]\right)$$

$$= j^{-1}\left(\left(D^m{}_i \frac{\partial}{\partial x^m}(D^n{}_k) - D^m{}_k \frac{\partial}{\partial x^m}(D^n{}_i)\right)\cdot\frac{\partial}{\partial x^n}\right)$$

= evaluate the input vector field at id.

⊛ $$\left[D^m{}_1 \frac{\partial}{\partial x^m}, D^n{}_2 \frac{\partial}{\partial x^n}\right] = \left[D^m{}_1 \frac{\partial}{\partial x^m}(D^n{}_2) - D^m{}_2 \frac{\partial}{\partial x^m}(D^n{}_1)\right]\cdot\frac{\partial}{\partial x^n}$$

since $D^n{}_1$ & $D^n{}_2$ are impt ; $D^1{}_2 = D^3{}_2 = 0$.

$$= -D^m_2 \frac{\partial}{\partial x^m}(D^1_1)\cdot\frac{\partial}{\partial x^1} + \left(D^m_1 \frac{\partial}{\partial x^m}(D^2_2) - D^m_2 \frac{\partial}{\partial x^m}(D^2_1)\right)\frac{\partial}{\partial x^2}$$

$$- D^m_2 \frac{\partial}{\partial x^m}(D^3_1)\frac{\partial}{\partial x^3}$$

$$= -D^1_2 \frac{\partial}{\partial x^1} + (D^1_1 + D^2_2)\frac{\partial}{\partial x^2} - D^3_2 \frac{\partial}{\partial x^3}$$

$$= 2x^1 \frac{\partial}{\partial x^2}$$

similarly,

$$\left[D^m_1 \frac{\partial}{\partial x^m}, D^n_3 \frac{\partial}{\partial x^n}\right] = -2x^2\frac{\partial}{\partial x^1} - 2\left(\frac{1+x^2x^3}{x^1}\right)\frac{\partial}{\partial x^3}$$

$$\left[D^m_2 \frac{\partial}{\partial x^m}, D^n_3 \frac{\partial}{\partial x^n}\right] = x^1\frac{\partial}{\partial x^1} - x^2\frac{\partial}{\partial x^2} + x^3\frac{\partial}{\partial x^3}$$

$$\Rightarrow \left[\!\left[\left(\frac{\partial}{\partial x^1}\right)_{id}, \left(\frac{\partial}{\partial x^2}\right)_{id}\right]\!\right] = 2\left(\frac{\partial}{\partial x^2}\right)_{id} \Longrightarrow \boxed{C^2_{12} = 2}$$

$$\left[\!\left[\left(\frac{\partial}{\partial x^1}\right)_{id}, \left(\frac{\partial}{\partial x^3}\right)_{id}\right]\!\right] = -2\left(\frac{\partial}{\partial x^3}\right)_{id} \Longrightarrow \boxed{C^3_{13} = -2}$$

$$\left[\left(\frac{\partial}{\partial x^2}\right)_{id}, \left(\frac{\partial}{\partial x^3}\right)_{id}\right] = \frac{\partial}{\partial x^1} \Longrightarrow \boxed{C^1_{23} = 1}$$

non-zero structure constants.